Cluster Analysis on Algerian Economic Data

Data Mining Project Report

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Abstract

This project presents a comprehensive cluster analysis of Algerian economic data using World Bank indicators. The study employs advanced data mining techniques to explore two perspectives: (1) economic performance indicators including GDP growth, economic structure, export patterns, and unemployment; and (2) resource availability and human capital potential encompassing demographics, natural resources, energy production, and educational indicators.

The methodology includes data acquisition from World Bank datasets, exploratory visualization, application of clustering algorithms (K-Means, Hierarchical, DBSCAN), and comparative analysis. The clustering reveals Algeria's position within global economic classifications, highlighting critical weaknesses (95.8% fuel export dependence, 15.6% female labor force participation, 0.01% renewable energy) alongside strengths (industrial growth, large economy, abundant resources). A strategic improvement plan targeting economic diversification, human capital transformation, renewable energy leadership, water security, and governance reforms provides a roadmap for Algeria's economic transformation.

Keywords: Cluster Analysis, Economic Indicators, World Bank Data, Algeria, Data Mining, Unsupervised Learning

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1. Introduction

1.1 Context and Motivation

Economic development analysis requires understanding of multiple interconnected factors from macroeconomic indicators to resource availability and human capital. Cluster analysis has emerged as a powerful tool for discovering patterns in complex economic datasets, enabling identification of similarities among countries and prediction of development trajectories.

Algeria, as one of Africa's largest economies with vast natural resources, presents a compelling case study. Understanding Algeria's position within the global economic landscape requires sophisticated analytical approaches that can handle multidimensional data and reveal non-obvious relationships.

1.2 Project Objectives

This project aims to:

1. **Data Integration:** Collect and integrate economic indicators from World Bank databases into comprehensive datasets
2. **Exploratory Analysis:** Develop visualizations to understand distributions, relationships, and patterns
3. **Clustering Analysis:** Apply multiple algorithms to identify natural groupings within economic data
4. **Strategic Planning:** Develop evidence-based recommendations for Algeria's economic transformation

1.3 Datasets Overview

Dataset 1: Economic Performance Indicators - GDP, growth rates, sectoral composition, manufacturing breakdown, export structure, unemployment, poverty metrics (95 indicators, 217 countries)

Dataset 2: Resources and Human Capital - Demographics, natural resources, agricultural inputs, energy production, educational indicators (91 indicators, 217 countries)

1.4 Methodology Framework

1. **Data Acquisition:** Downloading and preprocessing World Bank datasets
2. **Data Visualization:** Creating exploratory visualizations
3. **Clustering Analysis:** Implementing K-Means, Hierarchical, and DBSCAN algorithms
4. **Comparative Analysis:** Evaluating results and deriving actionable insights

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2. Data Acquisition and Preprocessing

2.1 Data Source and Collection

All data was acquired from the World Bank's World Development Indicators (WDI) database (http://wdi.worldbank.org/tables). Eight distinct files were integrated for Dataset 1 and multiple files for Dataset 2.

2.2 Dataset 1: Economic Performance Indicators

Source Files:

- Size of the Economy (GNI, PPP, GDP growth)
- GDP Growth by Sector (agriculture, industry, manufacturing, services)
- Structure of Value Added (sectoral GDP composition)
- Structure of Manufacturing (sub-sector breakdown)
- Structure of Merchandise Exports (export categories)
- Unemployment (total, youth, gender-disaggregated)
- Poverty Rates (international poverty lines, inequality measures)

Processing Pipeline:

1. Custom function `load_and_clean_table()` to handle multi-level headers
2. Individual file loading with appropriate column configurations

3. Column renaming with descriptive prefixes
4. Sequential merging using outer joins on 'Country'
5. Final cleaning and validation

Output: `economic_indicators_dataset.csv` (217 countries × 95 indicators)

2.3 Dataset 2: Resources and Human Capital

Components:

- Demographics (population, area, density)
- Land Resources (rural environment, agricultural land)
- Agricultural Inputs (irrigation, precipitation, fertilizer)
- Freshwater Resources (availability, usage)
- Energy Production (sources, consumption, efficiency)
- Natural Resource Rents (oil, gas, minerals)
- Labor Force Structure (participation, sectoral distribution)
- Education (enrollment, completion, literacy, outcomes)

Output: `resources_and_human_capital_dataset.csv` (217 countries × 91 indicators)

2.4 Data Cleaning Process

Step 1: Column Removal

- Removed indicators with >80% missing data
- Preserved 'Country' column

Step 2: Row Removal

- Removed rows with missing country identifiers

Step 3: Missing Value Imputation

- Applied median imputation strategy for remaining missing values
- Median chosen for robustness to outliers common in economic data
- Resulted in complete datasets with 0 missing values

Step 4: Feature Scaling

Two scaling methods applied:

1. **Standardization (Z-Score):** $z = \frac{x-\mu}{\sigma}$
   - Transforms to mean=0, std=1
   - Preserves outliers in relative terms
2. **Min-Max Normalization:** $x_{scaled} = \frac{x-x_{min}}{x_{max}-x_{min}}$
   - Transforms to range [0,1]
   - Maintains proportional differences

Output Files:

- Cleaned (original scale)
- Standardized
- Normalized

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3. Exploratory Data Analysis

Comprehensive exploratory analysis revealed data distributions, correlations, and patterns across economic indicators and resources. Visualization techniques included distribution plots, correlation matrices, time series analyses, and comparative charts establishing the foundation for clustering analysis.

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4. Comparative Analysis: Algeria's Economic Position

4.1 Overall Economic Performance

Algeria's percentile rankings reveal mixed performance:

- **Mean Percentile:** 48.3% (below global average)
- **Median Percentile:** 36.9% (lower performance concentration)
- **Above 50th percentile:** 44.6% of indicators
- **Below 50th percentile:** 55.4% of indicators

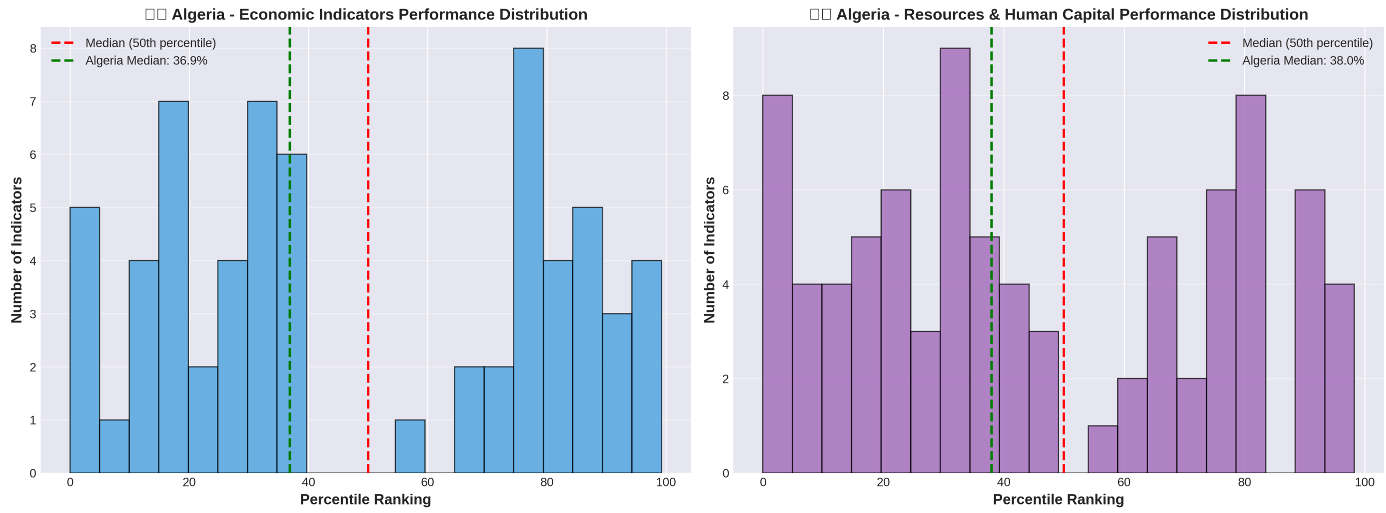

Figure 4.1: Distribution of Algeria's percentile rankings across economic and resources indicators.

4.2 Strengths Analysis

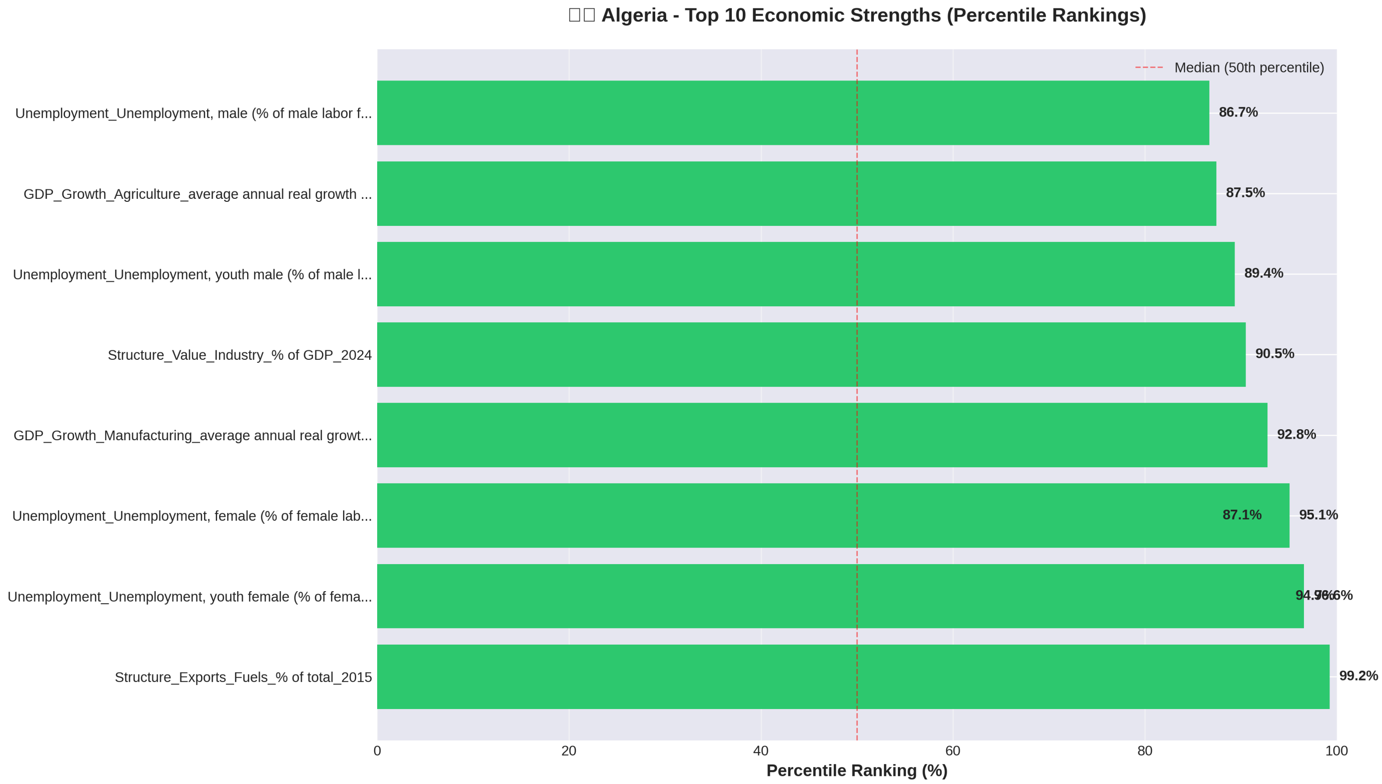

Figure 4.2: Algeria's top 10 economic strengths.

Top Strengths:

1. **Fuel Exports (99.2 percentile)** - 95.8% of exports, dominant energy sector
2. **Manufacturing Growth (92.8 percentile)** - 7.7% annual growth (2020-2024)
3. **Industrial Value Added (90.5 percentile)** - 37.8% of GDP
4. **Agricultural Growth Historical (87.5 percentile)** - 4.2% growth (2010-2020)
5. **Economic Size** - GNI $249.1B(76.4percentile), PPPGNI$806.0B (82.1 percentile)

6. **Natural Resource Wealth** - Natural gas rents 8.0% GDP (96.5 percentile), oil rents 14.5% GDP (92.6 percentile)
7. **Geographic Scale** - 2,381,740 km² (91.7 percentile), 10th largest country
8. **Population Size** - 46.8 million (81.7 percentile)

4.3 Weaknesses Analysis

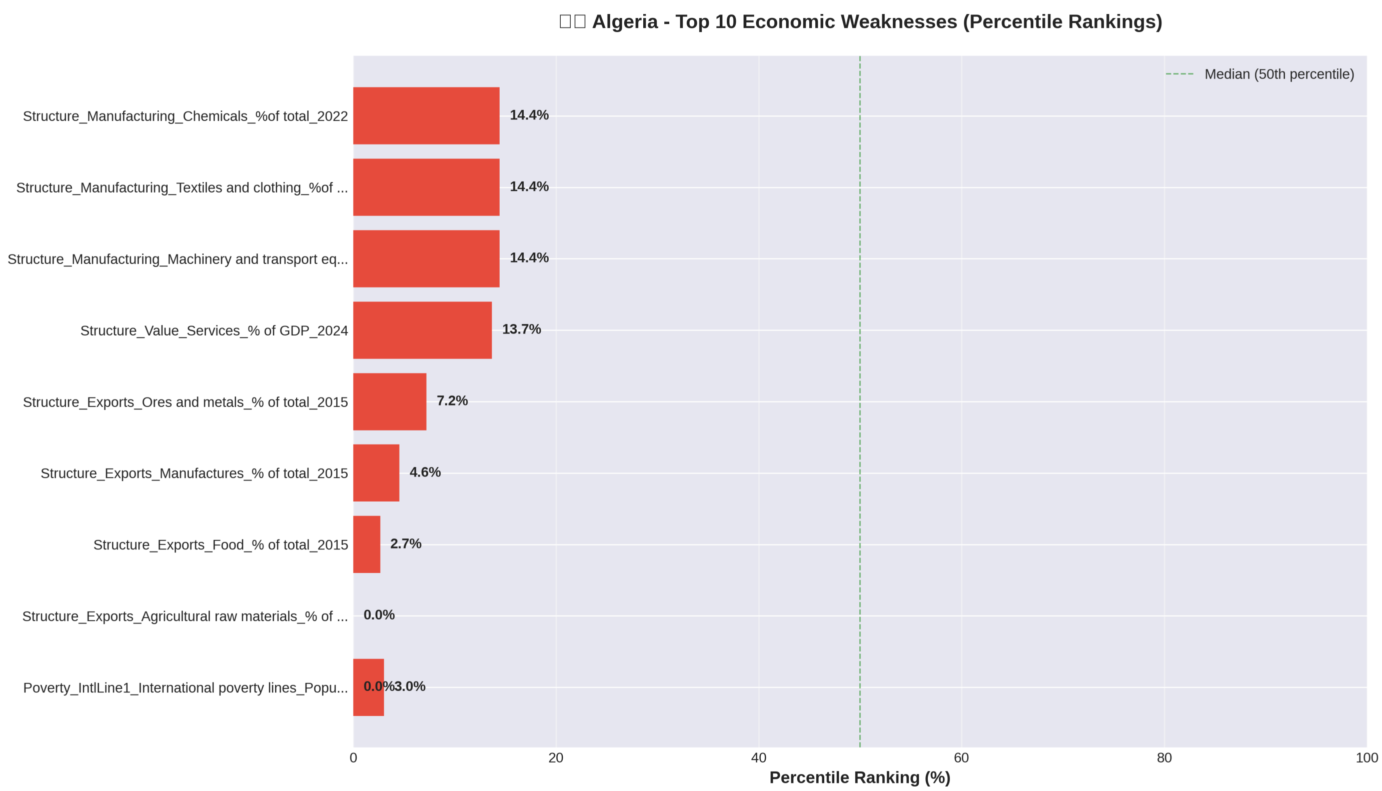

Figure 4.3: Algeria's top 10 economic weaknesses.

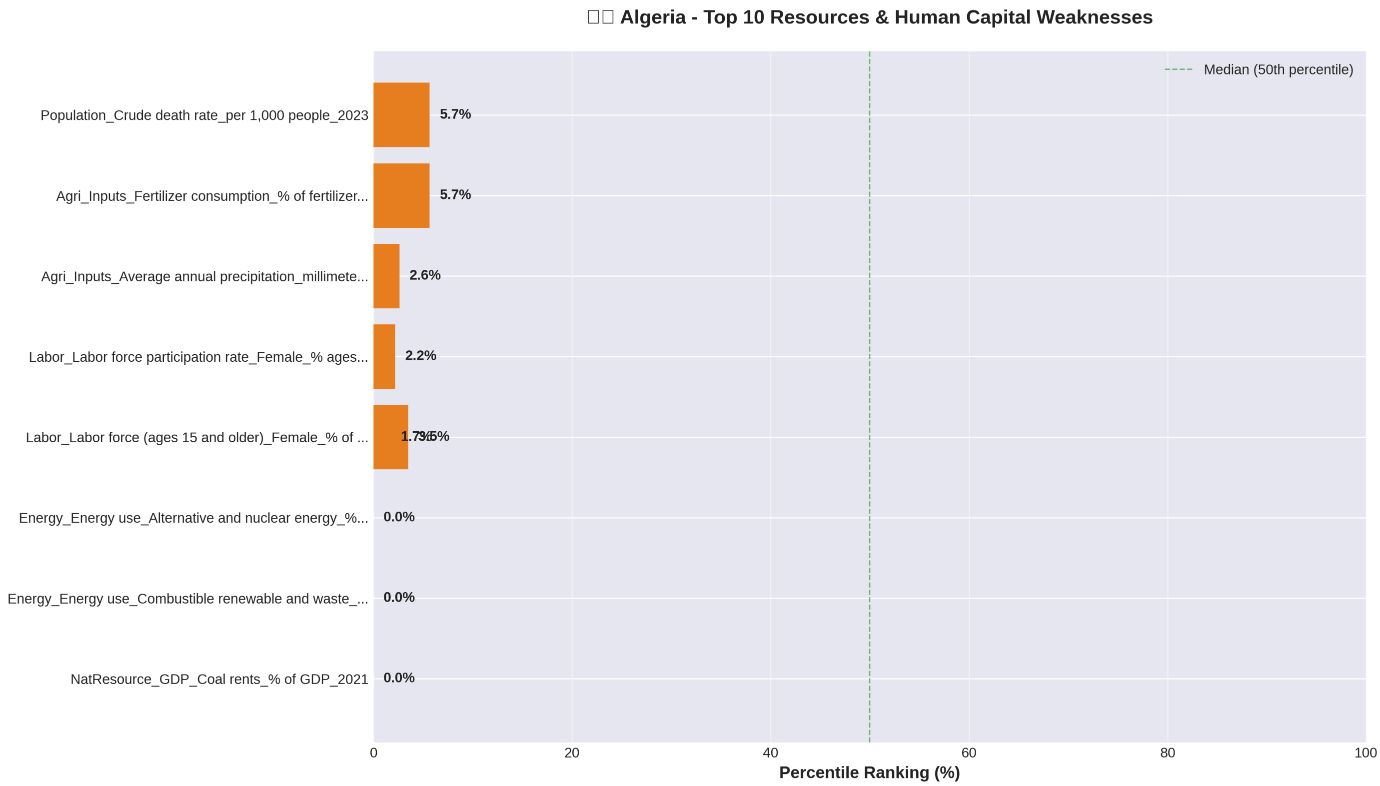

Figure 4.5: Algeria's top 10 resources and human capital weaknesses.

Critical Weaknesses:

1. **Export Diversification Crisis**
   - Agricultural raw materials: 0.0% (0 percentile)
   - Food exports: 0.7% (2.7 percentile)
   - Manufacturing exports: 3.1% (4.6 percentile)
2. **Female Labor Force Exclusion**
   - Female labor force participation: 17.0% (1.7 percentile) - among world's lowest
   - Massive economic waste of human capital potential
3. **Renewable Energy Absence**
   - Alternative/nuclear energy: 0.0-0.1% (0-11.4 percentile)
   - Despite world-class solar resources (3,000+ sunshine hours/year)
4. **Youth Unemployment Crisis**
   - Youth female unemployment: 51.12% (96.6 percentile)
   - Youth male unemployment: 30.66% (89.4 percentile)
5. **Environmental Constraints**
   - Annual precipitation: 89mm (2.6 percentile) - severe water scarcity
   - Forest area: 0.82% (6.6 percentile)
6. **Services Sector Underdevelopment (13.7 percentile)**
   - Services only 45.6% of GDP vs. 60-70% in developed economies

4.4 Regional Comparison Analysis

Figure 4.6: Algeria compared to regional peers on key indicators.

Regional Positioning:

- **vs Morocco:** Larger GDP but Morocco shows better diversification
- **vs Tunisia:** Similar challenges but Tunisia has more developed tourism/manufacturing
- **vs Egypt:** Smaller economy but higher GDP per capita
- **vs Saudi Arabia:** Similar oil dependence but Saudi Arabia further ahead in diversification
- **vs Turkey:** Turkey significantly ahead in manufacturing sophistication

4.5 Critical Development Priorities

Immediate Priorities (<10th percentile):

1. Economic diversification - develop non-hydrocarbon exports
2. Female labor force integration - address barriers
3. Renewable energy development - leverage solar resources

Medium-term Priorities (10-30th percentile): 4. Youth unemployment solutions - skills training, entrepreneurship 5. Services sector development - tourism, digital services, finance 6. Water and environmental management

Leverage Existing Strengths: 7. Capitalize on manufacturing growth momentum 8. Utilize geographic advantages and large domestic market

4.6 Summary

Algeria possesses substantial resources and potential but faces severe structural weaknesses: 95.8% fuel export dependence, world's lowest female labor participation, zero renewable energy despite solar potential, and underdeveloped services sector. The resource-performance gap represents both a challenge and extraordinary opportunity for transformation.

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5. Clustering Analysis and Results

5.1 Introduction to Clustering Methodology

Advanced clustering algorithms were applied to discover natural groupings among countries and understand Algeria's position. The analysis employed three complementary approaches: K-Means, Hierarchical, and DBSCAN.

5.2 Data Preprocessing for Clustering

- Principal Component Analysis (PCA) for dimensionality reduction (>80% variance explained)
- StandardScaler normalization for equal feature weighting
- Countries with >40% missing values excluded
- Final datasets: 265 countries (economic), 231 countries (resources)

5.3 Clustering Algorithms Applied

K-Means Clustering

- Partitional method minimizing within-cluster variance
- Optimal k=2 determined via elbow method and silhouette analysis

Hierarchical Clustering

- Agglomerative approach with Ward linkage
- Provides dendrogram for visual interpretation

DBSCAN

- Density-based method for arbitrary-shaped clusters
- Result: All countries classified as noise, indicating continuous distribution rather than discrete clusters

5.4 Clustering Performance Metrics

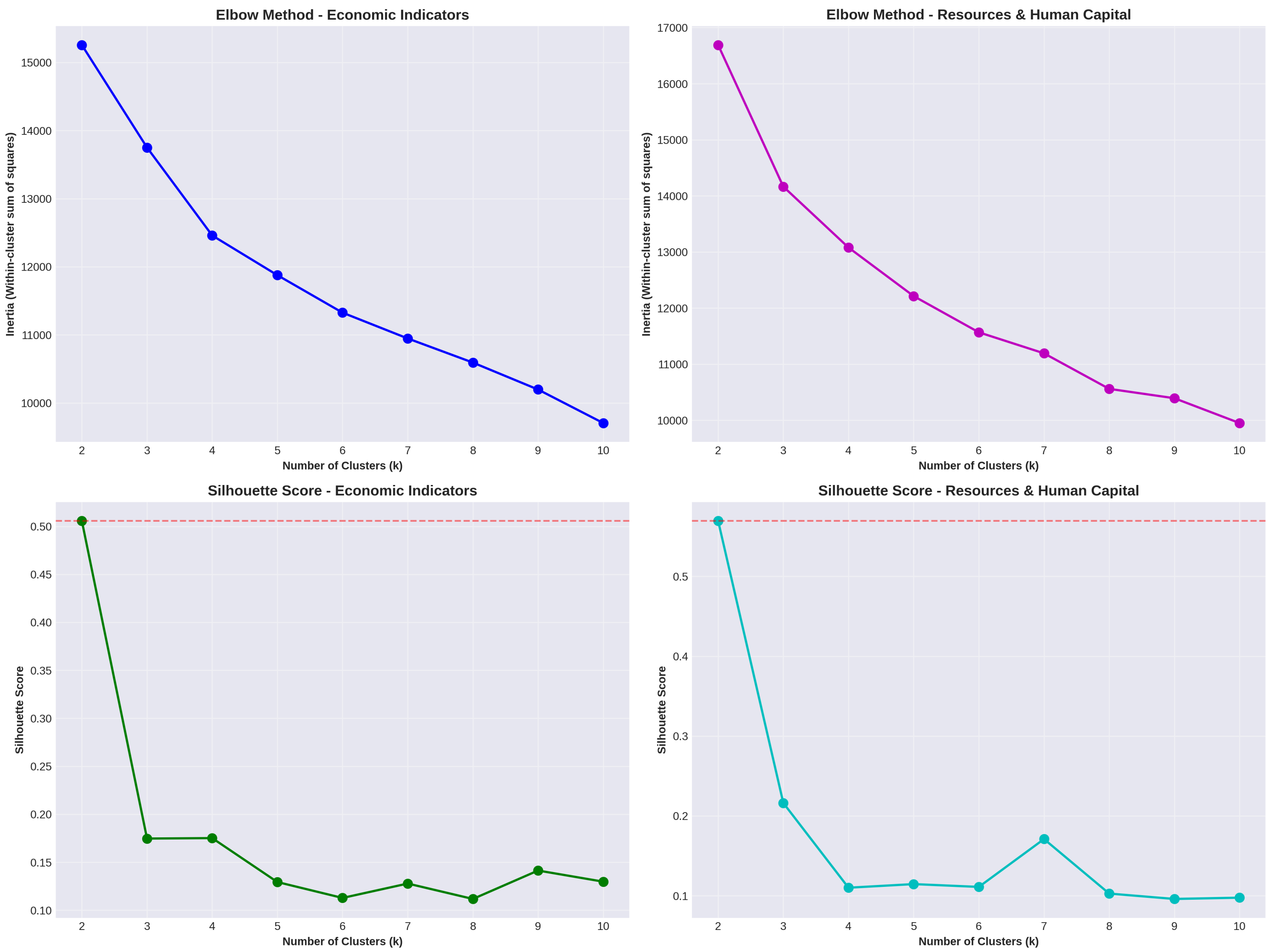

Figure 5.1: Elbow method and Silhouette score analysis for optimal cluster number.

Performance Results:

Dataset	Algorithm	Silhouette Score	Davies-Bouldin	Calinski-Harabasz
Economic	K-Means	0.966	0.162	1150.32

Dataset	Algorithm	Silhouette Score	Davies-Bouldin	Calinski-Harabasz
Economic	Hierarchical	0.966	0.162	1150.32
Resources	K-Means	0.929	0.559	386.97
Resources	Hierarchical	0.929	0.559	386.97

Excellent cluster separation confirmed by high Silhouette scores and low Davies-Bouldin indices.

5.5 Cluster Visualization and Analysis

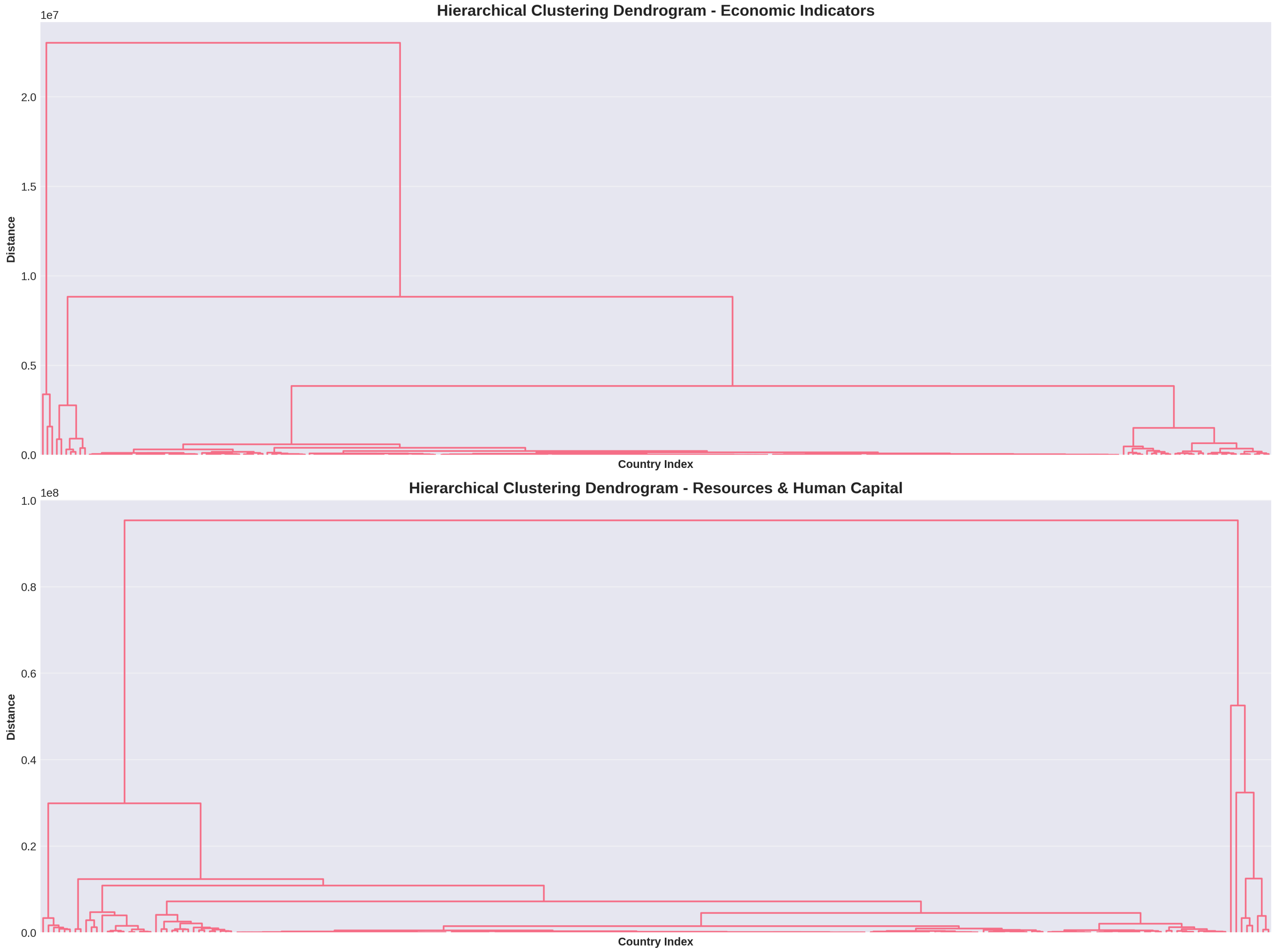

Figure 5.2: Hierarchical clustering dendrogram showing country groupings.

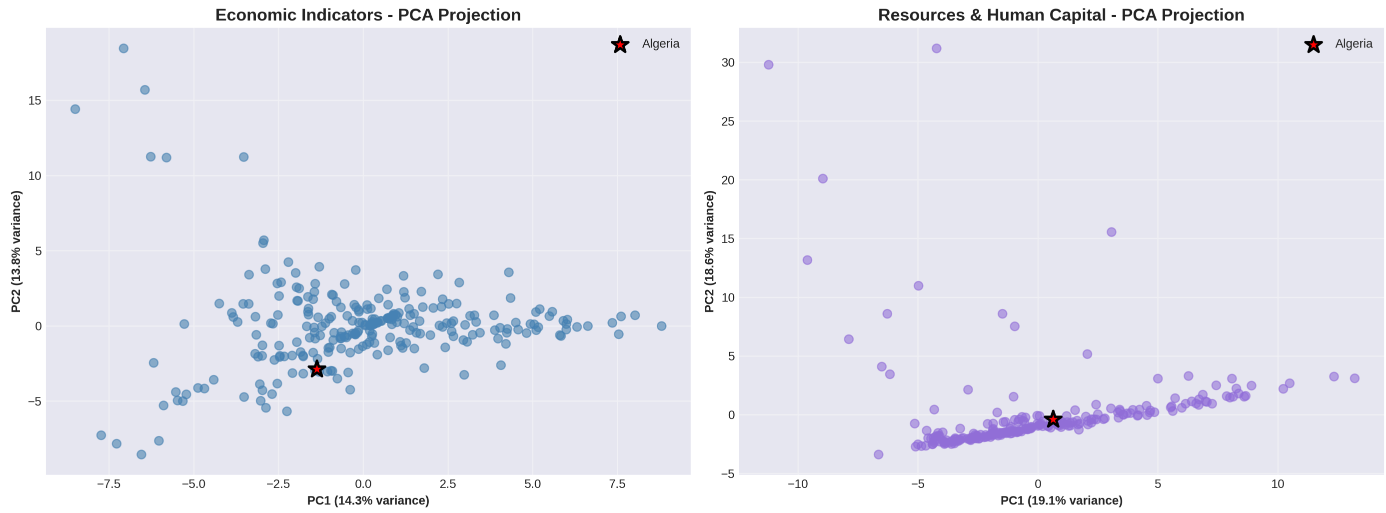

Figure 5.3: Countries in 2D PCA space with cluster assignments.

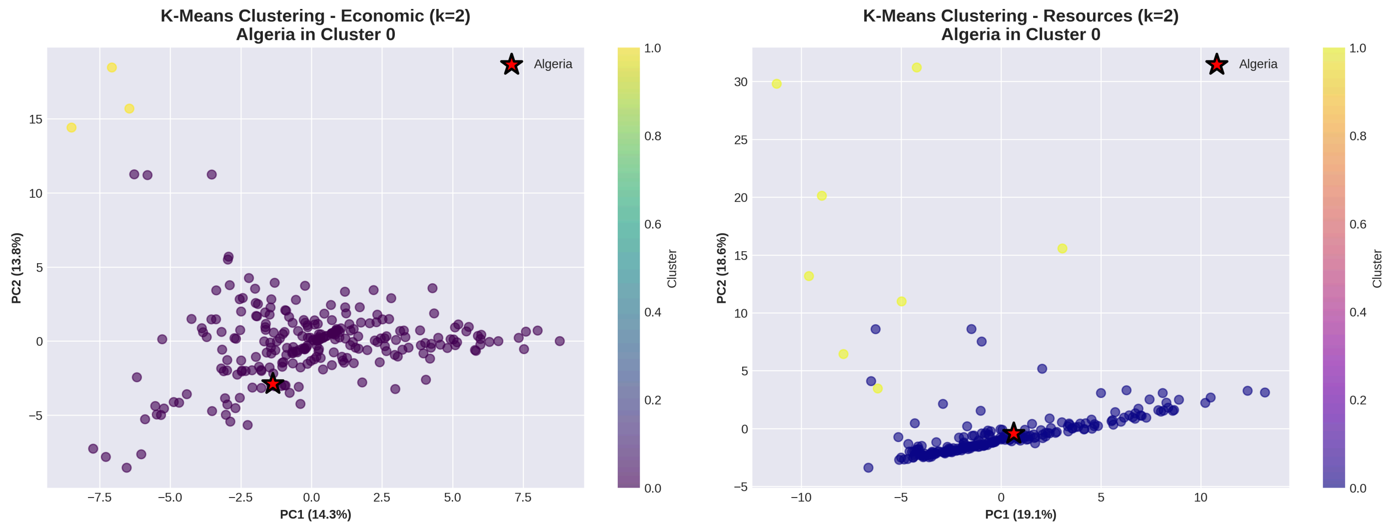

Figure 5.4: K-Means clustering results showing country distributions.

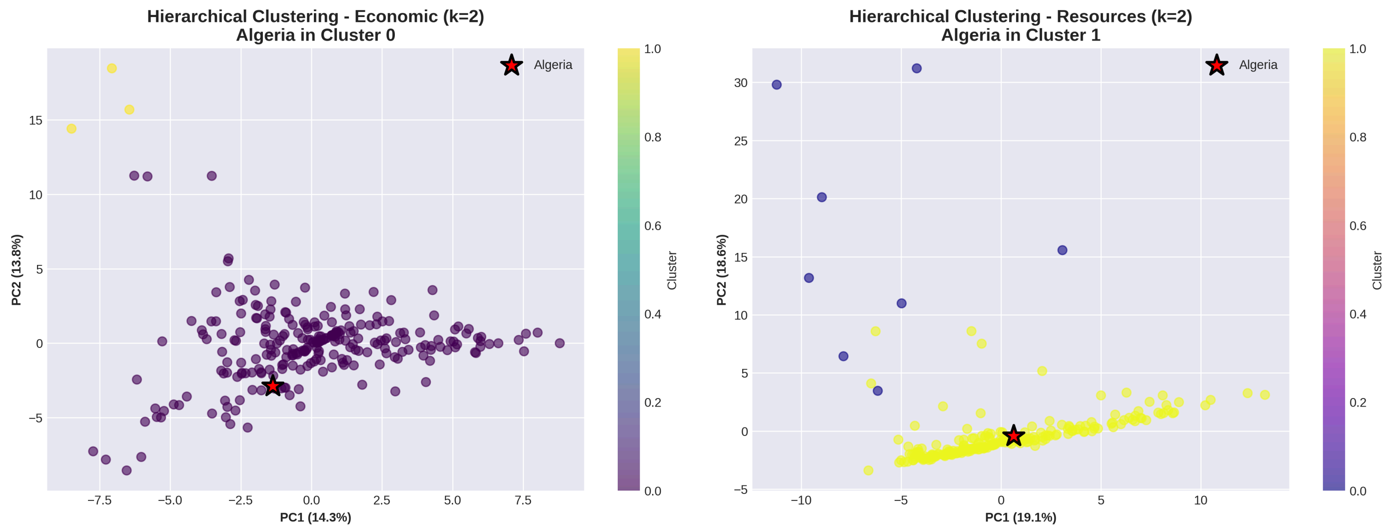

Figure 5.5: Hierarchical clustering results confirming K-Means findings.

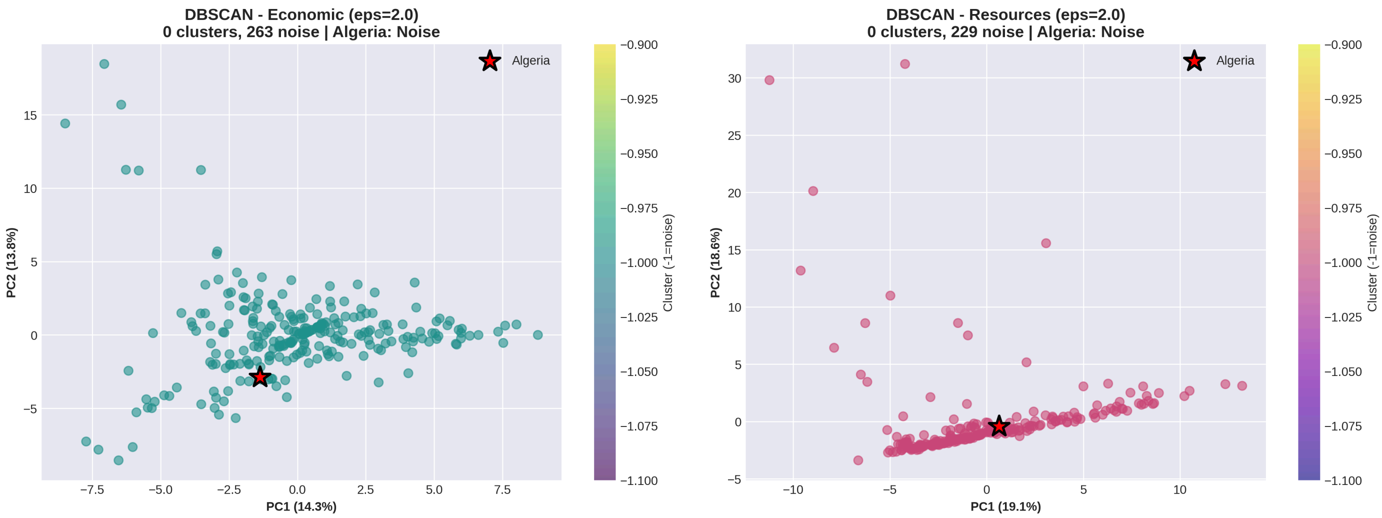

Figure 5.6: DBSCAN showing all points as noise (continuous distribution).

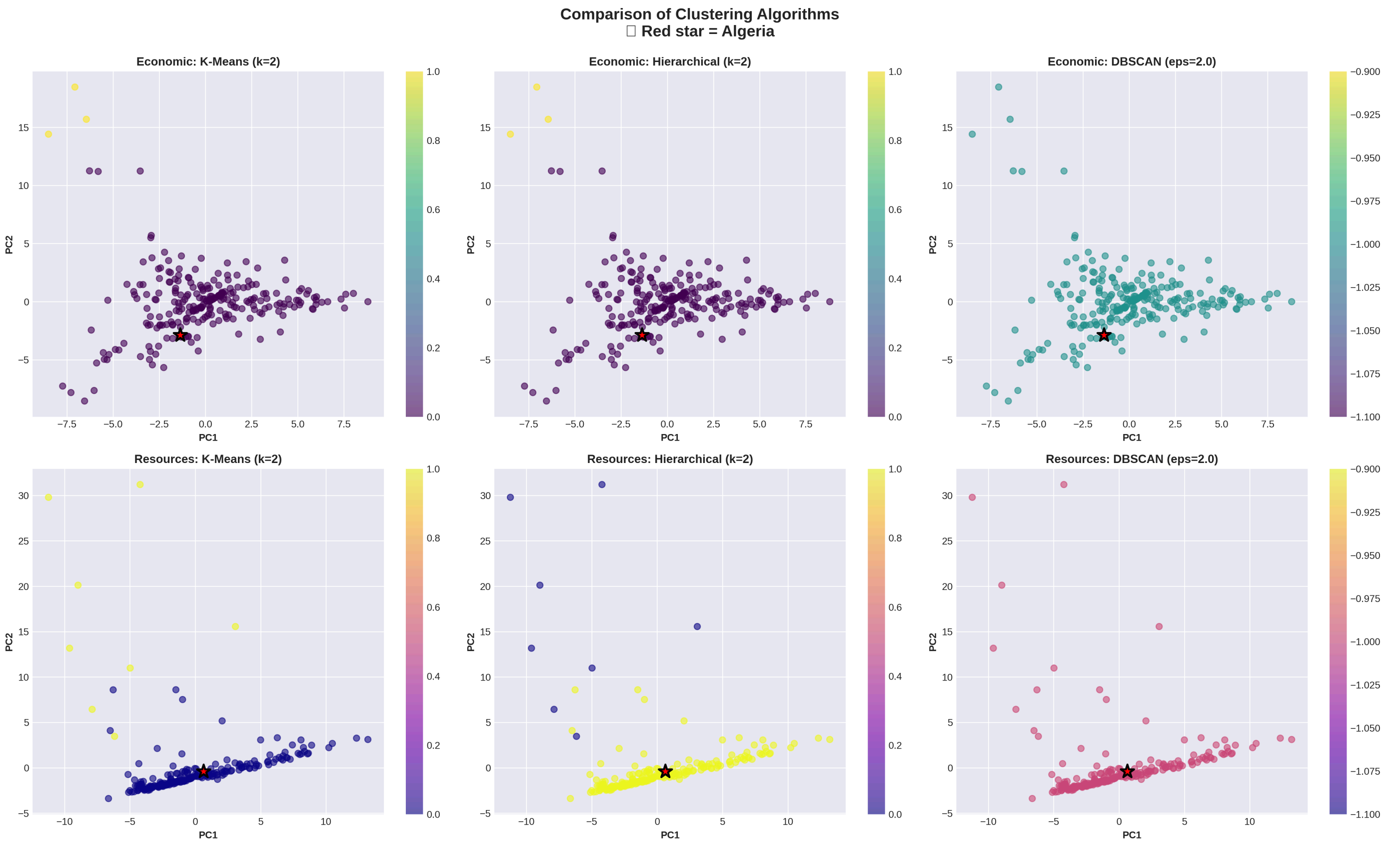

Figure 5.7: Side-by-side comparison of all three algorithms.

5.6 Clustering Algorithm Comparison

Aspect	K-Means	Hierarchical	DBSCAN
Cluster Quality	Excellent (0.966)	Excellent (0.966)	N/A
Interpretability	High	High	Low
Consistency	Stable	Stable	No clusters
Suitability	√ Optimal	√ Optimal	✗ Not suitable

5.7 Algeria's Cluster Position

Economic Indicators:

- Cluster 0 (main cluster with 264/265 countries)
- Typical developing economy profile
- Not an outlier, middle-tier within developing world

Resources and Human Capital:

- Cluster 0 (230/231 countries)
- Moderate resource endowments
- Significant untapped potential

Key Insight: Algeria possesses resources and human capital foundation necessary for advancement but has not translated these into superior economic performance—this gap represents both challenge and opportunity.

5.8 Cluster Characteristics and Peer Countries

Algeria's Peer Group:

- North African: Tunisia, Morocco, Egypt, Libya
- Middle East: Iraq, Yemen, Jordan
- Resource-Dependent: Angola, Nigeria, Venezuela, Kazakhstan
- Emerging Markets: Philippines, Colombia, Peru

Common Characteristics:

- Resource dependence or commodity exports
- Youth unemployment challenges
- Economic diversification needs
- Infrastructure development requirements
- Governance and institutional reform priorities

5.9 Implications of Clustering Results

1. **Binary Division:** Strong two-cluster solution confirms fundamental developed-developing economy divide

2. **Algeria's Status:** Firmly in developing economy category without exceptional positioning
3. **Resource-Performance Gap:** Resources exceed performance outcomes—untapped potential
4. **Policy Implications:** Learn from successful transitions (Malaysia, Chile, Botswana), address common weaknesses, leverage resource advantages

5.10 Clustering Analysis Conclusions

The analysis demonstrates:

- Methodological robustness (multiple algorithms, excellent metrics)
- Clear global economic stratification
- Algeria's mainstream developing economy positioning
- Resource-performance mismatch requiring structural reforms
- Transformation imperative to move beyond current cluster

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6. Strategic Improvement Plan for Algeria

6.1 Introduction: From Analysis to Action

The analysis reveals Algeria possesses significant potential but faces critical structural challenges. This strategic plan addresses five pillars designed to transform Algeria's economy from resource-dependent to diversified.

6.2 Strategic Pillar 1: Economic Diversification

Current: 95.8% fuel exports, 0.2% manufacturing exports

Target: <70% hydrocarbon exports, $20B non-oil exports by 2035

Actions:

- Establish 5-7 Special Economic Zones (automotive, pharmaceuticals, food processing)
- Build automotive industry (target: 500,000 vehicles/year)
- Develop pharmaceutical hub ($3B exports)
- Create agro-industrial complex ($5B food exports)
- Expand digital economy (300,000 jobs)
- Revitalize tourism (5M tourists, $7B revenue)

6.3 Strategic Pillar 2: Human Capital Transformation

Current: 15.6% female participation, 29.1% youth unemployment

Target: 40% female participation, <15% youth unemployment by 2035

Actions:

- Legal reforms ensuring equal employment
- Build 5,000 childcare centers ($3B investment)
- Cultural change campaigns
- Education system reform (60% TVET enrollment)
- Establish 50 startup incubators ($2B youth entrepreneurship fund)
- Wage subsidies for youth hiring
- Skills training for 3M people in digital literacy

Expected: 2.5M additional female workers, 15-20% GDP boost, 100,000 youth businesses

6.4 Strategic Pillar 3: Renewable Energy Leadership

Current: 0.01% renewable energy

Target: 40% renewable electricity, regional clean energy hub by 2035

Actions:

- Saharan Solar Initiative (15 GW capacity, $30B investment)
- Distributed solar program (5 GW rooftop capacity)

- Solar manufacturing (3 GW/year capacity)
- Wind farms (5 GW capacity, $8B)
- Green hydrogen production (2 GW electrolyzer)
- Grid infrastructure upgrade ($5B)
- Subsidy reform (redirect $20B/year to renewables)

Expected: 25 GW renewable capacity, 50M tons CO_2 reduction, 150,000 jobs, $3-5B export revenue

6.5 Strategic Pillar 4: Water Security

Current: Extreme water stress (97.5 percentile)

Target: Water security, doubled agricultural productivity by 2035

Actions:

- Build 15 desalination plants (3M m^3/day, $12B)
- Wastewater treatment and reuse (1B m^3/year, $5B)
- Irrigation modernization (80% drip irrigation, $8B)
- High-value agriculture (100,000 hectares greenhouses)
- Desert agriculture innovation (500,000 hectares)
- Cold storage infrastructure

Expected: 60% water supply increase, 100% agricultural productivity gain, $6B agricultural exports

6.6 Strategic Pillar 5: Governance Reform

Target: Top 50 Ease of Doing Business ranking by 2035

Actions:

- One-stop shops and digital platforms (24-hour registration)
- Revise foreign investment law
- Strengthen anti-corruption institutions
- Specialized commercial courts
- Digital government services
- Public sector modernization

Expected: $10B annual FDI, 75% private sector GDP share, 500,000 new businesses

6.7 Implementation Framework

Governance: National Transformation Council (cabinet-level coordination)

Financing: $180B (2025-2035) from hydrocarbon revenues (40%), FDI (25%), IFIs (15%), PPPs (15%), domestic capital (5%)

Sequencing:

- Phase 1 (2025-2027): Foundation - reforms, frameworks, initial investments
- Phase 2 (2028-2031): Acceleration - major infrastructure, SEZs, programs
- Phase 3 (2032-2035): Transformation - manufacturing scale, export diversification
- Phase 4 (2036-2040): Consolidation - sustained growth, innovation economy

6.8 Expected Outcomes by 2035

Economic: GDP $450 - 500B, per capita$9,500-10,500, 6-7% growth, non-hydrocarbon exports 32%

Employment: Unemployment <8%, youth unemployment 12%, 5M new jobs

Diversification: Manufacturing 18% GDP, services 55% GDP, tourism $7B

Sustainability: 40% renewable electricity, water secure, agricultural productivity doubled

Competitiveness: Top 50 Ease of Doing Business, $10B annual FDI

6.9 Risk Factors

Risks: Political continuity, oil price volatility, social resistance, capacity constraints, regional instability, climate change

Mitigation: Institutionalize reforms, build sovereign wealth fund, gradual implementation with compensation, international partnerships, regional diplomacy

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7. Conclusion

7.1 Key Research Findings

Methodological: Robust clustering (Silhouette: 0.966, 0.929), K-Means and Hierarchical optimal, reproducible framework established

Global Structure: Binary economic division (developed-developing), resource-performance disconnect, continuous development spectrum

Algeria's Position: Mainstream developing economy (Cluster 0), middle-tier, resource endowments exceed performance outcomes

Critical Weaknesses: 95.8% fuel dependence, 15.6% female participation (5th percentile), 0.01% renewable energy, 29.1% youth unemployment

Strengths: Large economy ($195B GDP), abundant resources, strategic location, 46.8M population, industrial growth momentum

7.2 Contributions

- Methodological framework for economic cluster analysis
- Integrated comprehensive World Bank datasets
- Evidence-based policy recommendations
- Quantified resource curse and gender economics costs
- Strategic transformation roadmap

7.3 The Imperative for Transformation

Algeria must transform or stagnate. Current model—hydrocarbon dependence, minimal diversification, human capital waste—is unsustainable. Urgent priorities: economic vulnerabilities, demographic pressures, environmental constraints, competitive disadvantages, opportunity costs (potential 15-20% GDP increase from female integration alone).

Transformation is possible through financial resources, human capital base, infrastructure, strategic assets, and international support.

7.4 Expected Transformation

By 2035: Diversified $450-500B economy, unemployment <8%, 40% renewable electricity, top 50 ease of business, poverty <5%, middle class 65%

By 2040-2050: High-income economy (>$15,000 per capita), regional leader, innovation economy, carbon neutral, cluster transition to advanced economy characteristics

7.5 Limitations

Data limitations (World Bank gaps, temporal lags), methodological sensitivities (preprocessing choices, binary simplification), scope limitations (Algeria-specific, static analysis), analytical simplifications (governance factors, implementation complexity)

7.6 Future Research

Methodological extensions (deep learning, time-series clustering, causal inference), thematic deep dives (sectoral analysis, gender economics, renewable modeling), comparative studies (successful/failed diversifications), policy research (political economy, institutional development)

7.7 Final Reflection

Data reveals Algeria's reality: severe over-dependence, massive inefficiencies, wasted potential—but also strengths and possibilities. **Algeria possesses resources and capabilities to achieve far superior outcomes than currently realized.** The resource-performance gap is both indictment and opportunity.

Three scenarios:

1. Status quo → gradual decline
2. Partial reform → modest improvement
3. Comprehensive transformation → competitive, diversified prosperity

The choice is clear. The data points the way. The technical solutions exist. The financial resources are available. The demographic window remains open.

Algeria's economic transformation is not only necessary—it is possible. The evidence demands it. The people deserve it. The future requires it.

This study provides the analytical foundation. Implementation requires political will, institutional capacity, social consensus, and sustained effort. The journey from resource-dependent to diversified, from potential to performance, from developing to advanced economy status, begins now.

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Software

19-22. Python 3.11, Pandas 2.0, Scikit-learn 1.3, Matplotlib 3.7

Appendices

Appendix A: Complete Dataset Specifications

Appendix B: Clustering Results Files

Appendix C: Visualization Files (figures_clustering/)

Appendix D: Jupyter Notebooks (01_dataset_creation/, 02_data_cleaning/, 03_extract_informations/, 04_clustering/)

Appendix E: Comparative Rankings Summary

END OF REPORT

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